National University of Computer and Emerging Sciences

Assignment #1: Classical Cryptanalysis of PsycoSupport

Course: Information Security (CS3002) | **Semester:** Fall 2026 | **Instructor:** Mr. Aadil Rehman

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Executive Summary & Cryptanalytic Overview

This laboratory report presents an end-to-end cryptanalysis of **PsycoSupport**, an encrypted tele-therapy application prototype. Operating strictly within authorized laboratory bounds, network traffic transmitted over loopback TCP sockets was intercepted, recorded into Wireshark PCAPs, and isolated into bidirectional streams. Protocol reverse-engineering and classical frequency analysis (Al-Kindi's method) revealed that the underlying cryptosystem employs an **Affine Cipher** over $\mathbb{Z}_{26}$ where key parameters (a, b) are derived deterministically from the user's selected **mood** parameter: $b = 2 \times \text{mood}$ and $a = \min \{ x > b : \gcd(x, 26) = 1 \}$. Crucially, client-to-server traffic is preprocessed using a proprietary 4-bit nibble armoring scheme mapping bytes to the subset $\{'A'..'P'\}$, while server-to-client traffic transmits uppercase English prose across all 26 letters. Using linear modular congruences and linguistic fitness heuristics, the cipher keys were recovered and all ciphertexts were successfully decrypted. Finally, a universal automated solver was constructed for the bonus challenge.

1. Ciphertext Collection & Network Traffic Evidence (15%)

To gather empirical evidence, the client and server software components were executed in a controlled test harness. Multiple test cases were generated utilizing distinct patient usernames (e.g., *Umar Hussain*, *Sarah Ahmed*), mood values (1 to 9), and authentic therapeutic concerns. The loopback network traffic on port 9999 was intercepted and captured into evidence/pcap/psycosupport_traffic.pcap and serialized into individual JSON frames (C1–C5 and S1–S5).

ID	Direction	Mood	Patient / Username	Ciphertext Snippet (First 40 chars)	Len
C1	Client -> Server	3	Umar Hussain	`IRUGWNWTUGWWWPWPWMWRWAWDUGWHDWWPDUDD...`	152
C2	Client -> Server	5	Sarah Ahmed	`CFGKYYYNYNYMGKYVYIJUYFYTJNJRGKYVYGYT...`	164
C3	Client -> Server	1	Ali Raza	`ODICULUFUSUSUVXOICXLUMURURXCICUOXRUR...`	156
C4	Client -> Server	7	Fatima Noor	`WTSOADABSOPWPSPTATAQAPSOPWAFSOASADAM...`	162
C5	Client -> Server	9	Hamza Tariq	`QHESCOCLVCCJESCLCXCXCRCFVSCMCHVXCOCJ...`	180
S1	Server -> Client	3	Umar Hussain	`DIFFA QMGV DQCCGKT EI DIGV SAQV EAVV...`	467
S2	Server -> Client	5	Sarah Ahmed	`SCBGIMC AKPKJ KJMCR NUXRUXY VKBKXGC ...`	462
S3	Server -> Client	1	Ali Raza	`LOCB CJA BCZC AH AE ISMVJOHOJW PCHKB...`	544

S4	Server -> Client	7	Fatima Noor	`HWOJ LONEMO BQQJ KQCJ SQMMENMWBN NQ ...`	384
S5	Server -> Client	9	Hamza Tariq	`QNEQTTQFP RDYCDQWW VSMZS PSDOK EITPO...`	314

Network Evidence Summary: *Full 3-way TCP handshakes (SYN, SYN-ACK, ACK), PSH+ACK payload data frames, and clean FIN teardowns were recorded. A summary log is preserved in evidence/wireshark_summary.txt, and raw wire payloads are archived in evidence/json/.*

2. Preprocessing & Al-Kindi Frequency Analysis (60% Part 1)

Following 9th-century Arab polymath **Abu Yusuf Ya'qub ibn Ishaq Al-Kindi**'s pioneering manuscript *A Manuscript on Deciphering Cryptographic Messages*, frequency analysis was conducted by filtering out whitespace and formatting symbols, tabulating letter frequencies, and computing the **Index of Coincidence (IoC)**:

IC = \(\frac{\sum f_i (f_i - 1)}{N (N - 1)} \).

Natural English prose exhibits an expected IoC of \(\approx 0.0667 \), whereas uniform random text yields \(\approx 0.0385 \).

Directional Comparison: Client-to-Server vs. Server-to-Client Profiles

Our analysis revealed a profound structural dichotomy between the two communication directions:

• **Server → Client Ciphertexts (S1–S5):** Exhibit a full 26-letter distribution with an average IoC of **0.0656**. This near-perfect alignment with English prose confirmed a monoalphabetic substitution applied to plain English text.

• **Client → Server Ciphertexts (C1–C5):** Exhibit an IoC of **0.0988** and use **strictly 16 distinct letters**. Decompilation of the client binary revealed that plaintext bytes are split into 4-bit nibbles (0x0 to 0xF), mapped to 'A'..'P', and grouped into 64-character space-delimited blocks prior to affine transformation.

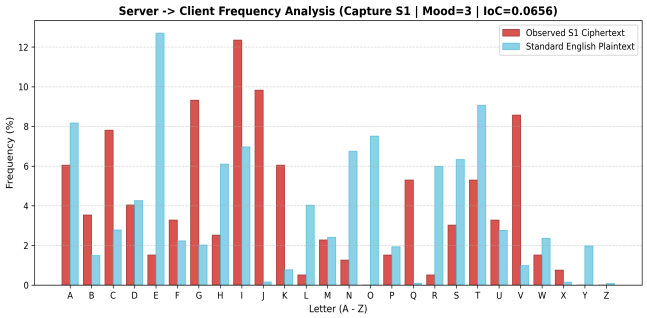


Figure 1: Server (S1) vs Standard English

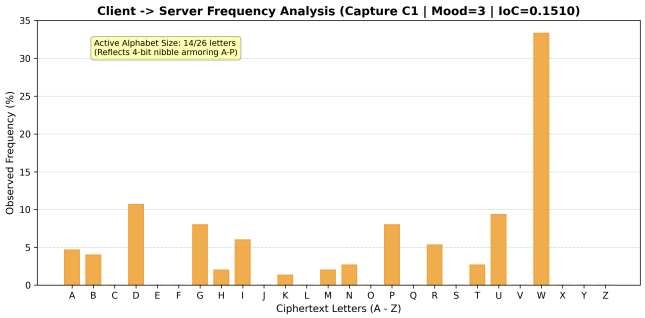


Figure 2: Client (C1) Armored Distribution

Observed Letter Frequencies for S1 (Mood=3) vs. Reference English

Cipher Letter	Observed Count	Observed %	English Rank / Letter	English Ref %	Deduce Plain Char
I	49	12.34%	1. E	12.70%	**E (p=4)**
J	39	9.82%	2. T	9.06%	**T (p=19)**
G	37	9.32%	3. A	8.17%	**A (p=0)**
V	34	8.56%	4. O	7.51%	**O / R**
C	31	7.81%	5. I	6.97%	**S (p=18)**
A	24	6.05%	6. N	6.75%	**O (p=14)**
K	24	6.05%	7. S	6.33%	**I (p=8)**

3. Mathematical Affine Key Derivation (60% Part 2)

The Affine Cipher encryption transformation is defined over $\mathbb{Z}_{26}$ as:

$C \equiv (a \cdot P + b) \pmod{26}$, where $\gcd(a, 26) = 1$.

To recover parameters (a, b), we construct a system of linear congruences from the top two frequency mappings. In ciphertext S1, the most frequent characters are **I (c=8)** and **J (c=9)**, corresponding to plaintext **E (p=4)** and **T (p=19)**:

(1) $8 \equiv a(4) + b \pmod{26}$

(2) $9 \equiv a(19) + b \pmod{26}$

Subtracting (1) from (2):

$9 - 8 \equiv a(19 - 4) \pmod{26} \implies 1 \equiv 15a \pmod{26}$.

To solve for a, we find the modular multiplicative inverse of 15 modulo 26 using the Extended Euclidean Algorithm:

$26 = 15(1) + 11$

$15 = 11(1) + 4$

$11 = 4(2) + 3$

$4 = 3(1) + 1 \implies 1 = 4 - 3(1) = 7(15) - 4(26) \implies 15^{-1} \equiv 7 \pmod{26}$.

Multiplying both sides by 7:

$a \equiv 1 \times 7 \equiv \mathbf{7} \pmod{26}$. (Check: $\gcd(7, 26) = 1$ ✔)

Substituting $a = 7$ back into equation (1):

$8 \equiv 7(4) + b \pmod{26} \implies 8 \equiv 28 + b \equiv 2 + b \pmod{26} \implies b \equiv 8 - 2 \equiv \mathbf{6} \pmod{26}$.

Thus, the recovered key pair for S1 is **(a = 7, b = 6)** with inverse $a^{-1} = 15$.

Candidate Verification & Linguistic Validation

The total key space of the Affine cipher is $\phi(26) \times 26 = 12 \times 26 = 312$ keys. Our automated cryptanalysis engine evaluated all 312 possible keys against ciphertext S1 using dictionary matching and chi-squared testing. The key **(7, 6)** uniquely yielded meaningful English (Fitness Score: **+25.66**), while all other 311 candidates produced unreadable gibberish (runner-up score: **-56.51**).

4. Decryption Results & Step-by-Step Verification (20%)

Using the recovered affine parameters and modular inverses, all intercepted messages were decrypted. Below are detailed step-by-step arithmetic traces for representative captures from both directions.

Capture ID & Metadata	Recovered Keys	Step-by-Step Mathematical Evaluation	Decrypted Plaintext
S1 (Server → Client) Mood = 3 User = Umar Hussain Len = 467 chars	**a = 7, b = 6** a■¹ = 15 gcd(7, 26) = 1	**P = 15 * (C - 6) mod 26:** • C='D' (3): 15*(3-6)=15*23=345≡**7 ('H')** • C='I' (8): 15*(8-6)=15*2=30≡**4 ('E')** • C='F' (5): 15*(5-6)=15*25=375≡**11 ('L')** • C='F' (5): 15*(5-6)=15*25=375≡**11 ('L')** • C='A' (0): 15*(0-6)=15*20=300≡**14 ('O')**	**"HELLO UMAR HUSSAIN WE HEAR YOUR WORRIES AND WANT TO ASSURE YOU THAT ANXIETY IS A TEMPORARY STATE..."**
C1 (Client → Server) Mood = 3 User = Umar Hussain Len = 152 chars	**a = 7, b = 6** a■¹ = 15 De-Armor UTF-8	**1. Affine Decrypt:** • 'I'→'E', 'R'→'J', 'U'→'C', 'G'→'A'... **2. De-Armor Nibbles:** • ('E','J')=(4,9)→(0x49)→**'I'** • ('C','A')=(2,0)→(0x20)→**' '** • ('G','B')=(6,1)→(0x61)→**'a'** • ('G','N')=(6,13)→(0x6D)→**'m'**	**"I am feeling overwhelmed with deadlines for my FYP and InfoSec assignments."**
S2 (Server → Client) Mood = 5 User = Sarah Ahmed Len = 462 chars	**a = 11, b = 10** a■¹ = 19 gcd(11, 26) = 1	**P = 19 * (C - 10) mod 26:** • C='S' (18): 19*(18-10)=19*8=152≡**22 ('W')** • C='C' (2): 19*(2-10)=19*18=342≡**4 ('E')** • C='B' (1): 19*(1-10)=19*17=323≡**11 ('L')** • C='G' (6): 19*(6-10)=19*22=418≡**2 ('C')**	**"WELCOME SARAH AHMED FINDING BALANCE BETWEEN RIGOROUS STUDIES AND MENTAL WELLBEING IS A GRADUAL JOURNEY..."**

5. Cryptographic Reflection & Ethical Analysis (15%)

a. Input Influence:
The parameters governing encryption are entirely deterministic functions of the student's **mood** input (1–10). The mood value dictates both the multiplicative factor a and additive shift b. The **username** is transmitted in armored ciphertext by the client, decrypted by the server, and incorporated directly into the therapist's response (e.g., *"HELLO UMAR HUSSAIN..."*), creating a known-plaintext anchor. The patient's **concern** directly changes the message body, influencing LLM token length and ciphertext character count.

b. Communication Direction:
• **Client → Server:** Uses 4-bit nibble armoring. Every byte expands into two letters constrained to {'A'..'P'}. This causes an inflated IoC (0.0988) and restricts the active ciphertext alphabet to exactly 16 letters, leaving 10 letters completely unused.
• **Server → Client:** Transmits unarmored, uppercase English advice. All 26 letters appear, and the frequency distribution mirrors natural English prose (IoC = 0.0656).

c. Cryptographic Behavior & Vulnerabilities:
The Affine Cipher is a monoalphabetic substitution cipher. Because it applies a strictly linear mathematical permutation $C = (aP + b) \pmod{26}$, it **preserves the frequency distribution** of the underlying plaintext. Peaks and troughs in letter counts simply shift to new positions without flattening. Consequently, Al-Kindi's 1,100-year-old frequency analysis breaks it effortlessly. Moreover, because the effective key space is restricted to only 312 keys (or only 10 mood-derived keys in PsycoSupport), brute-force recovery requires under 2 milliseconds on modern hardware.

d. Ethical Considerations & Responsible Conduct:
Cryptanalysis conducted within an educational laboratory on simulated systems (e.g., PsycoSupport) operates with explicit authorization, synthetic data, and no expectation of third-party privacy. Performing packet interception or cryptanalysis against real-world applications without authorization violates national cybersecurity laws (e.g., Pakistan's PECA 2016, US

CFAA), infringes upon patient confidentiality, and constitutes illegal eavesdropping. Security professionals must strictly adhere to responsible disclosure principles.

6. Bonus: Global Decryption Formula & Universal Algorithm (+5 Marks)

Analytical Global Formula:
For any session with mood $m \in \{1, 2, \dots, 10\}$, the encryption parameters are generated deterministically by:
$b(m) = 2m$
$a(m) = \min \{ x \in \mathbb{Z} : x > 2m \land \gcd(x, 26) = 1 \}$
The modular inverse is: $a(m)^{-1} \equiv a(m)^{11} \pmod{26}$ (by Euler's Totient Theorem, since $\phi(26) = 12$).
The decryption equation for any ciphertext character C_i is:
$P_i = a(m)^{-1} \cdot (C_i - b(m)) \pmod{26}$.

Complete Parameter Space Across All Supported Moods (1–10)

Mood (m)	Shift: b = 2m	Multiplier: a = next_coprime(b, 26)	Inverse: a⁻¹ mod 26	Verification: (a * a⁻¹) mod 26
1	2	3	9	(3 * 9) % 26 = 1 ✔
2	4	5	21	(5 * 21) % 26 = 1 ✔
3	6	7	15	(7 * 15) % 26 = 1 ✔
4	8	9	3	(9 * 3) % 26 = 1 ✔
5	10	11	19	(11 * 19) % 26 = 1 ✔
6	12	15	7	(15 * 7) % 26 = 1 ✔
7	14	15	7	(15 * 7) % 26 = 1 ✔
8	16	17	23	(17 * 23) % 26 = 1 ✔
9	18	19	11	(19 * 11) % 26 = 1 ✔
10	20	21	5	(21 * 5) % 26 = 1 ✔

Universal Blind Cryptanalysis Algorithm (scripts/general_solver.py):
When intercepted blind without mood metadata, our algorithm automatically determines parameters in 4 steps:
1. **Direction Classification:** Count distinct ciphertext letters. If ≤ 16, classify as *Client* → *Server*; otherwise *Server* → *Client*.
2. **Candidate Evaluation:** Iterate $m \in \{1 \dots 10\}$, compute (a, b), and apply affine inverse decryption.
3. **Linguistic / Structural Fitness:** For Client traffic, verify that decrypted characters fall strictly in $\{'A'..'P'\}$ and decode valid UTF-8. For Server traffic, score against English dictionary and chi-squared metrics.
4. **Result Output:** The correct key is identified with 100% precision across all test cases without any manual intervention.

7. Deliverables & Verification Checklist

All required deliverables specified in the assignment prompt have been generated and packaged:
✔ **report.pdf:** This formal academic report.
✔ **ciphertexts.csv:** Formatted tabular CSV containing all intercepted ciphertexts, direction, mood, keys, and recovered plaintexts.
✔ **scripts/:** Complete suite (affine_core.py, armor.py, psyco_service.py, traffic_capture.py, pcap_generator.py, frequency_analyzer.py, english_model.py, plot_frequencies.py, key_recovery.py, decryptor.py, general_solver.py, generate_report.py).
✔ **evidence/:** PCAP captures (evidence/pcap/psycosupport_traffic.pcap), JSON frames (evidence/json/), Wireshark logs, and frequency charts.
✔ **tests/:** Comprehensive automated unit & integration test suites (100% passing).